

Evaluation of LLM-driven Data Preparation

Abstract

Large language models (LLMs) have increasingly been used in data preparation pipelines due to their strong natural language understanding and reasoning capabilities. Data preparation tasks such as entity matching, and missing value imputation often rely on identifying syntactic patterns in the data, and domain-specific, contextual interpretations, the latter being a differentiating strength of LLM-based systems. However, these systems are imperfect demonstrating biases and inaccuracies. Studies have shown that 42% of model-centric approaches lead to inaccuracies and hallucinations, and 40% of models suffer from biases due to incomplete data. Existing work has studied LLM-based data preparation under fixed prompts and parameter configurations. In this paper, we present a systematic, comparative evaluation of LLM-driven entity matching, missing value imputation, and semantic reasoning across four state-of-the-art models. We evaluate the impact of various parameters including training example quantity, class ratios, input record lengths, missing values proportions, and the utility of key attribute values. Our results show the sensitivities and performance of these models using a suite of real data collections.

ACM Reference Format:

. 2026. Evaluation of LLM-driven Data Preparation. In *Proceedings of The 35th International ACM Conference on Knowledge and Information Management (CIKM '26)*. ACM, New York, NY, USA, 11 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Modern organizations increasingly rely on data preparation and data cleaning pipelines to integrate information collected from multiple heterogeneous sources. These processes are critical in domains such as healthcare, finance, scientific databases, and enterprise systems, where downstream decisions depend heavily on the quality and consistency of structured data. Common tasks in data preparation include identifying duplicated entities via entity matching, imputing missing values, and standardizing semantically related records across different systems. Recently, Large Language Models (LLMs) and Generative AI systems have been increasingly adopted for these tasks due to their strong natural language understanding and reasoning capabilities. Compared with traditional rule-based or statistical approaches, LLMs can interpret contextual information,

and generalize across heterogeneous schemas with limited manual engineering effort [2, 16]. As a result, LLM-based systems are now being explored for data integration, entity matching, missing value imputing, and for incorporating semantic context. However, these systems are imperfect, demonstrating performance limitations. Studies have shown that 42% of model-centric approaches lead to data inaccuracies and hallucinations, and 40% of models suffer from biases due to incomplete data [4]. Despite organizations spending an average 13% of their revenue in AI and ML model development over the next two years, an estimated average revenue loss of \$406M/year is expected due to under-performing AI models due to poor quality data [4].

The above studies underscore the importance of trustworthy and accurate performance of LLMs in data preparation tasks. Despite their strong generative and reasoning capabilities, LLMs often fail to reliably consider semantic context. While model outputs appear plausible, they often violate task-specific constraints or fail to include larger context. These limitations become particularly problematic in high-stakes domains such as healthcare, where errors may propagate through downstream systems. Consider the following motivating example showing the existing limitations of LLM usage in data preparation.

Example 1.1. Figure 1 illustrates three data cleaning tasks (entity matching (EM), missing value data imputation (DI), semantic reasoning (SR)), involving augmented dataset from the Mimic IV dataset [9]. We augment the records with attributes such as (synthetic) patient names, addresses, and contact information, while preserving real clinical attributes such as prescribed medications.

In entity matching (denoted in green), r_1, r_2 denote two patient records that must be reconciled. An LLM-driven solution indicates a positive classification (same entity). However, the ground truth indicates different persons where the middle name (*Wei* vs. *Li*) differentiates a male vs. female gender in r_1, r_2 , respectively. This naming convention commonly occurs in Asian countries; and the LLM lacks this cultural context. In data imputation (denoted in orange), records r_3, r_4 describe similar patient address and contact details but with missing City value in r_4 . An LLM imputes a City value that relies on address, health provider, while ignoring correlated attributes such as Phone area code and City. Lastly, in semantic reasoning (denoted in purple), can LLMs identify semantic equivalencies? Records $r_1 - r_6$ show various prescription drugs. To identify whether these records correspond to the same prescription class, LLMs such as GPT4o incorrectly classify r_4 and r_5 as equivalent, by overlooking key attributes such as Active Ingredients, Strength, Dosage, and over emphasizing on Drug name.

The above example demonstrates that existing generative models still face inherent limitations to apply context, and to identify differentiating attributes, when performing data cleaning tasks. To deepen LLM integration into the data preparation pipeline requires moving beyond the reliance of identifying syntactic patterns.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CIKM '26, Rome, Italy

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-XXXX-X/26/11
<https://doi.org/XXXXXXX.XXXXXXX>

ID	FName	LName	MName	Sex	YOB	Phone	City	Addr.	Provider	Drug	Active Ing.	Strength	Dosage	Route	Formulation
r_1	Zhang	Fang	Wei	—	1985	365-884-1274	Somerville	75 Main St	Tufts Medical Center	Ibuprofen	Ibuprofen	200mg	q6h	Oral	Tablet
r_2	Zhang	Fang	Li	F	1985	—/721/6384	—	78 Main St	Tufts Medical Center	Advil	Ibuprofen	200mg	q6h	Oral	Tablet
r_3	Johnson	Robert	NAN	M	1964	(617)555-8123	Boston	125 Washington St	Cambridge Health Alliance	Motrin IB	Ibuprofen USP	400mg	q8h	Oral	Tablet
r_4	Patel	Anika	R	F	1988	8572161397	—	78 Washington St	Cambridge Health Alliance	IBU-IV	Ibuprofen	800mg	q12h	Oral	Tablet
r_5	Garcia	Elena	M	F	1990	—	Quincy	51 Hancock St	Tufts Medical Center	IBU-IV	Ibuprofen	800mg	Continuous	Intravenous	Injection
r_6	Nguyen	Linh	NAN	M	1995	(781) 665-2309	Cambridge	18 Pearl St	Beth Israel Deaconess Medical Center	Apo-Ibuprofen	Ibuprofen USP	200mg	PRN	Oral	Capsule

1 Entity Matching	2 Data Imputation	3 Semantic Reasoning
Query: Are r_1, r_2 the same patient? Model: Yes, same patient. Truth: No, r_2 is a female patient. Limitation: Lack cultural gender cues; “Weifang” is more likely to be a male name.	Query: What is the value of City in r_4 ? Model: Boston. Truth: Cambridge. (857) is an area code around Cambridge. Limitation: High record similarity causes the model to overlook the area code.	Query: Cluster $\{r_1, \dots, r_6\}$ for the same drug. Model: $\{r_1, r_2\}, \{r_4, r_5\}$. Truth: $\{r_1, r_2\}$. All attributes except Drug are key attributes. Limitation: Model overly relies on drug name to differentiate.

Figure 1: Motivating example.

State-of-the-Art. Prior work have studied how to formulate entity matching, error detection, and data imputation as text generation tasks [14], refining and adapting instruction-tuned LLMs for data preprocessing [25], and evaluating ChatGPT-style models for entity matching under different prompting strategies [17]. Follow-on work has shown that including context and semantics into data cleaning considerably improves data repair accuracy [10, 13]. Existing work has motivated the need for context and semantics in data preparation, showing their impact on task-level performance, system, and prompt design. LLM-driven data preparation studies such as Jellyfish [25] further studies local instruction-tuned LLMs, covering multiple data preparation tasks while reducing dependence on closed-source APIs. These studies establish that LLMs can be used for data preparation tasks, but do not systematically explore the sensitivity of these models under varying parameter and data configurations. There remains a critical gap in understanding the limitations of LLMs under limited and varying: (i) context, controlled by the number of in-context examples and class-balance ratios; and (ii) semantics, controlled by missing values, retained record length, and partial key attributes.

In this paper, we aim to close this gap by performing a systematic evaluation of LLM-driven data preparation tasks across four foundation models (GPT-5.2, GPT-4o, Gemini 2.5 Pro, Gemini 2.5 Flash) on real data collections. Our evaluation studies three tasks: entity matching (EM), data imputation (DI), and semantic reasoning (SR); under varying data and parameter configurations, including the number of training examples, class balance ratios, input record lengths, missing value rates, and the utility of key attributes.

Contributions. We make the following contributions.

- (1) We perform a systematic, comparative evaluation of LLM-driven data preparation across varying data and parameter configurations, including the number of examples, class balance ratios, utility of key attributes, missing value rates.
- (2) We introduce a systematic evaluation framework that: (i) selects and curates training examples for each data preparation task (entity matching, imputation, semantic reasoning); (ii) query prompt construction and adaptation that refines subsequent prompts to gauge LLM logic and reasoning. Our adaptive prompt refinement

revealed that many LLMs rely on inherent learned rules to perform entity matching (e.g., similarity among specific attribute values), and cross-attribute correlations (e.g., inferring a missing city from phone area code or hospital name) for data imputation.

(3) Our evaluation reveal distinct model characteristics. For example, GPT models exhibit sensitivity to explicit rules, achieving high precision under key-attribute constraints, but at the cost of lower recall. In contrast, Gemini models demonstrate superior semantic reasoning, and maintain high recall under ambiguous and noisy settings. We report our preliminary, comparative results using real data collections and parameter configurations.

2 Related Work

LLMs for data preparation.: Recent work has explored large language models as general-purpose tools for data cleaning and data integration. Narayan et al. [14] formulate entity matching, error detection, schema matching, data transformation, and data imputation as text generation tasks, and show that foundation models can achieve strong zero-shot and few-shot results. Jellyfish [25] further studies data preprocessing with local instruction-tuned LLMs, covering multiple data preparation tasks while reducing dependence on closed-source APIs. These studies provide an initial glimpse of LLMs in the data preparation pipeline, but do not provide a comprehensive evaluation and analysis across varying parameter settings and semantic reasoning capabilities. Our work builds on this line of research, but shifts the focus from task-level performance to controlled behavior analysis. Instead of treating the prompt as a fixed input, we vary specific prompt and data factors, including the number of in-context examples, the positive-case ratio, the retained record length, and key-attribute settings. This allows us to measure how each factor changes precision, recall, F1, and imputation accuracy. Therefore, our study complements existing LLM-based data preparation systems by explaining which input conditions make LLM predictions more stable or more error-prone.

LLM-based entity matching.: Entity matching is a central task in data integration, and recent studies have evaluated LLMs for matching entity records. Peeters and Bizer [17] study ChatGPT for entity

matching and show that prompt design, in-context examples, and explicit matching rules can improve matching quality. Their later work [18] expands the evaluation to stronger closed-source and open-source models under zero-shot, few-shot, and fine-tuning settings. However, existing LLM-based entity matching studies mainly compare models, prompting strategies, or training settings. They do not isolate how individual prompt and data factors change the model’s matching behavior. In particular, we study how parameter settings alter the LLM decision boundary.

Data imputation and context-aware repair. Missing-value imputation and data repair also depend heavily on context. Huang and Hao [8] propose PRPMI, which uses trainable prompts and retrieval augmentation to improve missing-value imputation with pre-trained language models. Hayat et al. [7] introduce CRILM, which uses language models to generate contextually relevant descriptors for missing values. Earlier data cleaning systems such as Baran [13] also show that correction quality depends on value-level, tuple-level, and column-level context. These works demonstrate that richer context can improve imputation and repair quality.

Rather than adding retrieval, training, or external repair signals, we examine the scope of an LLM reasoning capability from the given input data and examples. This provides insights on the model’s internal reasoning behavior from external evidence. We vary the number of missing attributes, and the error rate to evaluate conditions when the models are able to recover masked values, and the parameter settings when reasoning begins to degrade.

Semantic reasoning over structured data. Sui et al. [20] show that LLM performance on tabular data tasks are sensitive to input format, record ordering, and prompt structure. In recent work on data discovery, Starmie [6] learns contextualized column representations for table union search, while Chorus [10] applies foundation models to data discovery and exploration tasks such as table classification, column annotation, and join-column prediction. These existing lines of work strongly motivate the need for semantics in data discovery, and study the problem of efficient semantic representations towards data discovery tasks.

Our semantic-reasoning evaluation differs from table representation learning and data discovery. We focus on identifying parameter and data configurations where the reliance on rule-based, matching approaches (which often use syntactic similarity) begin to transition to learning and applying semantic equivalencies. For example, in our drug-product setting, records are semantically and medically related, and are often grouped together when task-defined key attributes match, such as active ingredients, strength, dosage form, and route. We evaluate whether LLMs can distinguish broad semantic equivalencies from task-level requirements. By varying the number of key attributes, our experiments expose cases where the model relies on broad semantic equivalence instead of a task-specific matching rule.

3 Tasks, Models and Datasets

We provide background on the three data preparation tasks, followed by the foundation models, and datasets used in this paper.

Problem Setup. We consider entity matching (EM), data imputation (DI), and semantic reasoning (SR) and describe each of these tasks. Let D be a tabular dataset with N records, where each record contains m attributes denoted as $A_1 \dots A_m$. For each record $r_i \in D$, we denote the value of attribute A in record r_i as $r_i[A]$.

Entity Matching. Entity matching determines whether two records refer to the same real-world entity, i.e., given $r_1, r_2 \in D$, do the records represent the same entity? Entity matching has been studied extensively over the past decade using blocking techniques, classification, and declarative-based methods [14, 17, 18].

Data Imputation. Data imputation aims to infer missing (or null) values in D . The inferred values may be within the attribute domain or not. Existing work have proposed using statistical inference [12], clustering of similar values [21], rule-based methods that leverage attribute semantics and relationships [3], ML-based methods [19].

Semantic Reasoning. Data preparation tasks often require domain-specific interpretations that convey attribute value semantics and context. Trained LLMs are now able to provide this context via attribute names and their domain values. However, LLMs still face limitations to prioritize and identify attributes where semantic reasoning and context are needed.

Foundation Models. We evaluate four LLMs: GPT-5.2, GPT-4o, Gemini 2.5 Pro and Gemini 2.5 Flash. We select these models because our study focuses on prompt-based data preparation under realistic API usage, where users often choose between stronger but more expensive models and faster models with lower inference cost. Since these models are closed-source systems, their exact model sizes and training data details are not publicly disclosed.

GPT-5.2: GPT-5.2 is a frontier reasoning model released by OpenAI in December 2025. It belongs to the GPT family of autoregressive transformer-based language models and is designed for professional tasks that require multi-step reasoning, long-context understanding, coding, tool use, and vision-based analysis. Compared with earlier GPT models, GPT-5.2 places stronger emphasis on reliable reasoning and long-document understanding. We select GPT-5.2 as a representative high-capability reasoning model to evaluate whether stronger reasoning ability improves rule following, attribute comparison, and robustness in data preparation tasks. Its exact parameter count has not been publicly disclosed.

GPT-4o: GPT-4o is a multimodal “omni” model released by OpenAI in May 2024. It is an autoregressive model trained end-to-end across text, image, audio, and video, allowing the same model to process and generate information across different modalities. In this paper, we use GPT-4o for text-based structured-data tasks, but its design is still relevant because it emphasizes fast response, broad context understanding, and flexible instruction following. We include GPT-4o as a strong previous-generation OpenAI baseline and compare it with GPT-5.2 to study whether newer reasoning models improve

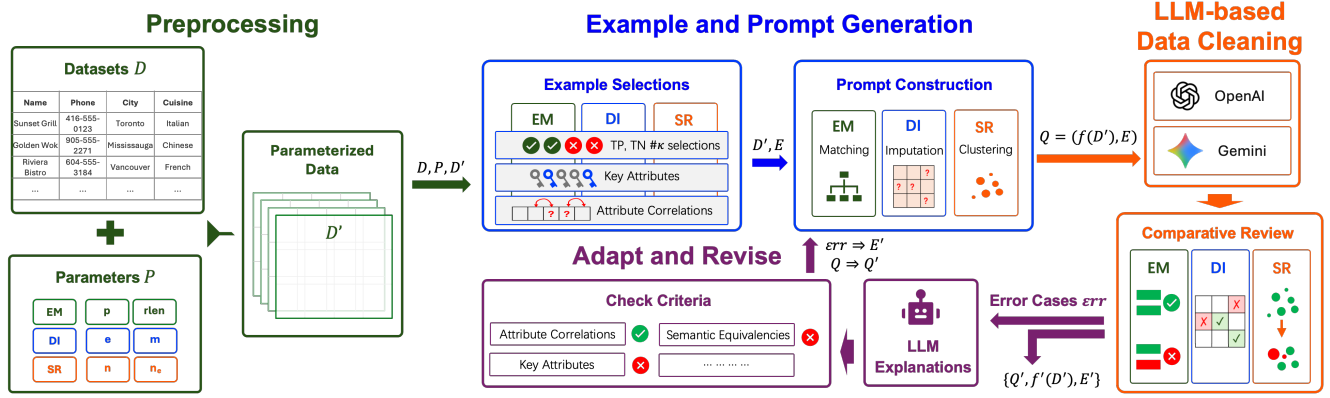


Figure 2: LLM-driven data preparation framework.

data preparation performance. Its exact parameter count has not been publicly disclosed.

Gemini 2.5 Pro: Gemini 2.5 Pro is a Google DeepMind reasoning model introduced in March 2025 as part of the Gemini 2.5 family. It is designed as a native multimodal model with a long context window and stronger reasoning ability than earlier Gemini models. Its long-context capability makes it suitable for tasks that require integrating evidence across many attributes, examples, and records. We select Gemini 2.5 Pro as a representative long-context reasoning model to test whether stronger context integration helps entity matching, data imputation, and semantic reasoning over structured records. Its exact parameter count has not been publicly disclosed.

Gemini 2.5 Flash: Gemini 2.5 Flash is a faster and more cost-efficient model in the Gemini 2.5 family, released in 2025. It is designed to balance reasoning quality, response latency, and inference cost, and supports a configurable thinking budget that lets users adjust the amount of reasoning computation used during inference. This makes it useful for data preparation pipelines where many records must be processed and cost can become a practical constraint. We include Gemini 2.5 Flash to evaluate the trade-off between efficiency and robustness, and to compare it with Gemini 2.5 Pro within the same model family. Its exact parameter count has not been publicly disclosed.

Datasets. We use the following real datasets:

iTunes – Amazon [5]: describing song name, artist, album, genre, price, release details, containing 539 records and 8 attributes.

Amazon – Google [5]: an integration of product records describing product names, manufacturer, pricing and product text description containing 4589 records with varying naming conventions, technical specifications and missing information.

Restaurant [11]: Fodor and Zagat guides containing 864 records describing restaurant name, address, city, phone and category.

Adult [22]: benchmark census data with approximately 50K records, describing demographic and employment attributes.

Drug [1]: an integration of prescription-drug records constructed from Orange Book [24], IDD [23], and RxNorm [15]. Orange Book provides FDA-approved drug products and therapeutic equivalence

information, IDD provides international drug-name and product records, and RxNorm provides normalized clinical drug names with ingredients, strengths, and dosage forms. The data contains 1000 drugs and 20 attributes describing drug properties.

4 LLMs for Data Preparation

Figure 2 shows our LLM-driven data prep evaluation framework consisting of four primary modules: (i) preprocessing, (ii) example and prompt generation, (iii) LLM-based data cleaning, (iv) adapt and revise. Given tabular data D , and configuration parameters P for each task, data preprocessing generates new versions of D , by filtering D according to varying parameter values to generate test data D' . For example, in entity matching, we evaluate the impact of class balance ratios (between true positives (TP) vs. true negatives (TN)) on classification accuracy, and vary the proportion of TP pairs from 20% to 100% in D' . Table 1 summarizes the list of parameters for each task, and their range of values, with defaults in bold.

Example Selection: Given the original data D , test data D' with labels L' , and parameters P , the *Example and Prompt Generation* module curates a set of input examples E using D and P , and subsequently constructs task-specific queries Q using E and D' . We randomly select a set of κ examples consisting of TP and TN cases for each task (i.e., consisting of true/false duplicates, missing values of varying error rates, and semantically (non-) equivalent entities) to construct an example set E . We further refine E to ensure characteristic features for model learning exist, such as key attributes, attribute correlations (e.g., city and phone area code), and minimum record length requirements. The example set size $|E|$ varies, from zero-shot to few-shot learning, where we formally evaluate the latter in our controlled experiments.

Prompt construction: Given E , D' , the *Prompt Construction* module creates task-specific prompt queries Q as follows:

- (a) EM: given two records $\{r_1, r_2\} \in D'$, and an equivalence function $f(r_1, r_2)$, Q poses a binary query of whether f returns true (r_1, r_2 refer to the same entity); or false (r_1, r_2 refer to distinct entities).
- (b) DI: given record $r_1 \in D'$ with missing values $\{r_1[A_1] \dots r_1[A_j]\}$,

Table 1: Parameters and defaults.

Task	Symbol	Description	Values
EM	s	# training examples	[0, 10] default 5
	p	% of TP (to TN) cases	[20, 100], default 50
	k	# key attributes	[1, 5] default 3
	r_{len}	% original record length	20, 40, 60, 80, 100
DI	m	# missing attributes	[1, 5] default 1
	e	% error rate	5, 10 , 15, 20, 25, 30
	s	# training examples	[0, 10] default 5
SR	n	# records	200, 400, 600 , 800, 1000
	n_e	# entities	4, 8, 12 , 16, 20
	p	% of TP (to TN) cases	15, 30, 45, 60 , 75, 100

for attributes $A_1 \dots A_j$, and an imputation function $f(r_1, A_1 \dots A_j)$. Query Q asks for the return (imputed) value(s) of $f = \{r_1^*[A_1] \dots r_1^*[A_j]\}$. (c) SR: given records $\{r_1 \dots r_j\} \in D'$, and a clustering function $f(r_1 \dots r_j) = \{C_i\}_{i=1}^N$, Q asks for a clustering of the records such that each cluster C_i represents a distinct entity.

LLM-based Data Cleaning evaluates the task-specific functions f described above. Specifically, we use foundation models to compute function f . For entity matching, f is evaluated based on classification accuracy, whether the pair satisfies the true positive or true negative ground truth. For data imputation, we evaluate model correctness if the imputed value(s) are correct, i.e., $\{r_1^*[A_1] = r_1^*[A_1] \dots r_1^*[A_j] = r_1^*[A_j]\}$, for ground truth values $\{r_1^*[A_1] \dots r_1^*[A_j]\}$. For semantic reasoning, we evaluate the model's ability to detect semantic equivalence among varying attribute values, and whether the entity e_i represented by each cluster C_i is correct, i.e., $e_i = e_i^*$, where e_i^* denotes the ground truth, representative entity.

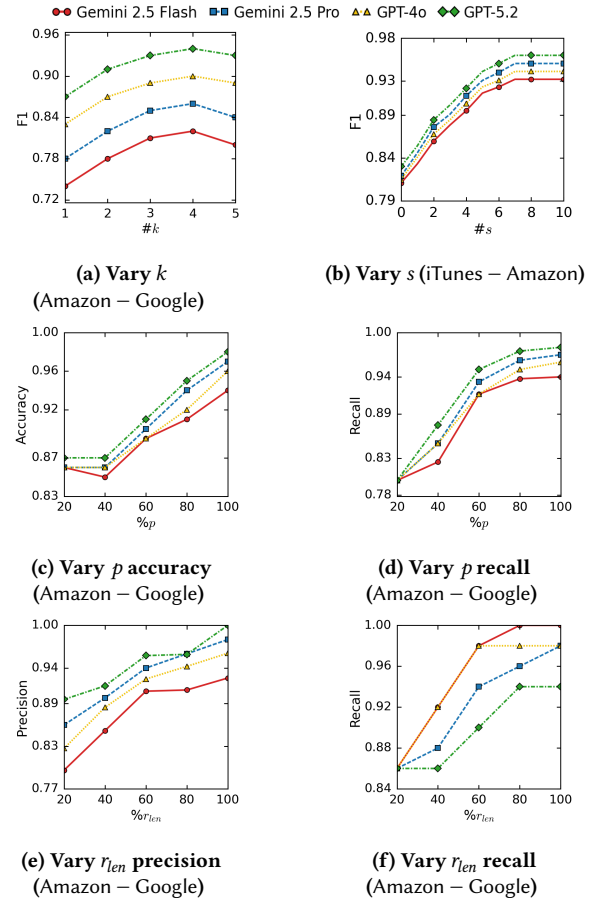
Given test labels L' , we compute the precision, recall, accuracy, and F1-score for each task-specific output of f . If any of these metrics falls below 95%, we identify the error cases $\mathcal{E}rr$. In the *Adapt and Revise* module, we evaluate the records in $\mathcal{E}rr$ along the following criteria to determine the error cause: (1) did the foundation model M utilize attribute correlations that exist in D' to compute f ? (2) were key attributes used to uniquely identify entities? (3) were attribute semantics and equivalencies used? We find the answers to these questions by asking M for explanations of how f was computed. In *Check Criteria*, we review the explanations, and check against our attribute correlations, key attributes, and semantic equivalencies criteria. If M did not consider these criteria, we revise $\mathcal{E}rr$ to explicitly include key and correlated attributes, and additional (similar) examples (generating a new E'), with a revised query prompt Q' that focuses on this subset of cases. We iterate until the model output $f'(D')$ achieves our threshold accuracy.

5 Experiments

Setup. We use the standard APIs for each model, and evaluate each model independently without conversation history. We evaluate the four LLMs using the task-to-dataset mapping shown in Table 2. Our code, and data are publicly available [1].

Table 2: Datasets and task mapping.

Dataset	EM				DI		SR			
	s	p	k	r_{len}	m	e	s	n	n_e	p
iTunes-Amazon	✓	✓	✓	✓	✓	✓	✓	–	–	–
Restaurant	✓	✓	✓	✓	✓	✓	✓	–	–	–
Amazon-Google	✓	✓	✓	✓	–	–	–	–	–	–
Adult	–	–	–	–	✓	✓	✓	–	–	–
Drug	–	–	–	–	–	–	–	✓	✓	✓

**Figure 3: Entity matching: LLM performance under different parameter settings.**

5.1 Entity Matching Performance

Exp-1: EM Vary k : Figure 3a shows the F1 score for varying number of key attributes. We observe that GPT-5.2 achieves an average +6% gain in F1 over the next closest baseline GPT-4. As the number of key attributes increase, we achieve an increasing F1 where more keys help to differentiate entities.

Exp-2: EM Vary s : Figure 3b shows that all models exhibit an average +20% in F1, from zero-shot to $s = 10$ examples. We observe that once $s = 6$, all models exhibit minimal performance gains. Entity

(song) variations such as remixes, 2.0 versions and re-releases, are often differentiated via these additional examples.

Exp-3: EM Vary p : Figure 3c and Figure 3d show the accuracy and recall performance, respectively, for the Amazon – Google dataset. We observe that as the percentage of TP p increases, model performance: (1) initially degrades in accuracy (for $30\% \leq p \leq 50\%$), where the added examples cause model confusion. However, for $p \geq 60\%$, we observe strong linear increases in accuracy across all models. Similar behaviour is seen in recall performance, with notable gains until $p = 60\%$, and marginal gains thereafter.

Exp-4: EM Vary r_{len} : 3e and Figure 3f show the precision and recall performance, respectively, for varying record length using the Amazon – Google dataset. As expected, shorter record lengths lead to lower scores as missing key attribute values facilitate entity resolution. We note that Gemini 2.5 outperforms all models by $\text{tb}\%$ in recall showing it’s robustness to missing values.

5.2 Data Imputation Performance

Exp-5: DI Vary e : Figure 4a shows decreasing F1 for increasing error rate, as expected. GPT models exhibit more robustness in the presence of noise than Gemini models. All models exhibit a dramatic linear decline as e approaches 30%.

Exp-6: DI Vary m : Figure 4b shows F1 decline across all models as the number of attributes with missing values increase. Again, the GPT models demonstrate superior robustness achieving an approximate +13% in F1 over the Gemini models.

Exp-7: DI Vary s : Figure 4c shows F1 scores for varying number of examples, s , on the iTunes – Amazon data. We observe that all models exhibit a linear increase in F1, with an approximate increase of +10% for GPT-4o and Gemini 2.5 Pro, and +18% for GPT-5.2 and Gemini 2.5 Flash, when moving from zero-shot to $s = 10$. We observe the largest F1 gains at approximately $s = 6$, with an average gain of less than 5% for each increment of s .

5.3 Semantic Reasoning Performance

Exp-8: SR Vary n_e : For semantic reasoning, Figure 5a shows that F1 gradually declines as the number of entities increase, making it more difficult to differentiate entities. All models achieve a notable 85% baseline F1, even at $n_e = 20$. For a smaller number of entities, all models achieve superior 90%+ F1-performance.

Exp-9: SR Vary n : Figure 5b shows the F1 scores for an increasing number of records in D' using the Drug dataset. We observe that having more records moderately degrades F1 performance (within 5% for each model). This is likely due to the increasing heterogeneity among the record formats, making it difficult to differentiate similar drugs. For example, differing drug names (brand vs. generic), active ingredients, dosage, routes and drug strength, for an increasing number of records, require models to understand semantic equivalence among the attribute values.

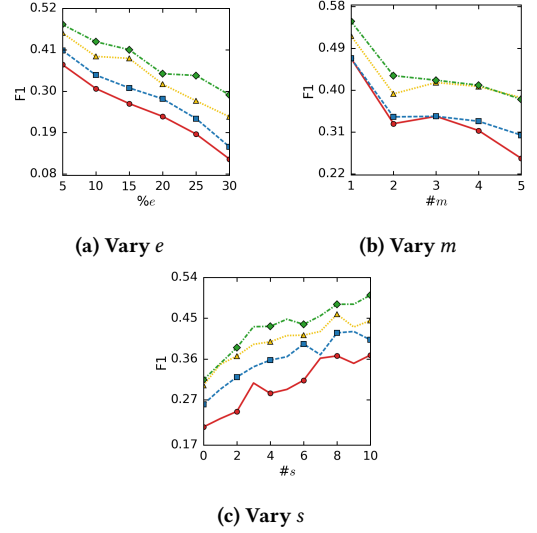


Figure 4: Data imputation performance (iTunes – Amazon).

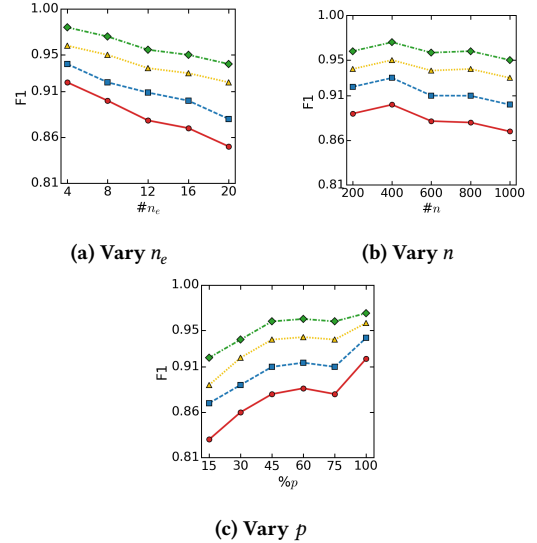


Figure 5: Semantic reasoning model performance (Drug).

Exp-10: SR Vary p : Figure 5c shows increasing F1 scores as the number of true positive examples increase. Models tend to be biased towards positive classification for increasing p . GPT models outperform Gemini models, on average, between 3-10% in F1-score. However, the Gemini models demonstrate greater benefit from more examples, particularly as $p > 75\%$, with larger incremental F1 gains.

6 Conclusion

We introduced a systematic evaluation of LLM-driven data preparation for entity matching, data imputation and semantic reasoning across four foundation models with varying parameter and data configurations. Our results showed the sensitivity of these models and their comparative performance. As next steps, we are studying metrics to evaluate model output coherence and consistency.

GenAI Usage Disclosure

Generative AI tools and technologies were not used to create content in this work.

References

- [1] Anonymous. 2026. Anonymous repository for Evaluation of LLM-driven Data Preparation. <https://anonymous.4open.science/r/Evaluation-of-LLM-driven-Data-Preparation-2B23/>.
- [2] Ioannis Arvanitis-Kasinikos et al. 2025. Entity Matching with 7B LLMs: A Study on Prompting Strategies. In *CEUR Workshop*.
- [3] Philip Bohannon, Wenfei Fan, Floris Geerts, Xibei Jia, and Anastasios Kementsis-sidis. 2007. Conditional Functional Dependencies for Data Cleaning. In *Proceedings of the 23rd International Conference on Data Engineering*. 746–755. <https://doi.org/10.1109/ICDE.2007.367920>
- [4] Vanson Bourne. 2024. AI in 2024 – hopes and hurdles, Insights Report. *Five Tran* (2024).
- [5] Sanjib Das, AnHai Doan, Paul Suganthan G. C., Chaitanya Gokhale, Pradap Konda, Yash Govind, and Derek Paulsen. [n. d.]. The Magellan Data Repository. <https://sites.google.com/site/anhaidgroup/projects/data>.
- [6] Grace Fan, Jin Wang, Yuliang Li, Dan Zhang, and Renée J. Miller. 2023. Semantics-aware Dataset Discovery from Data Lakes with Contextualized Column-based Representation Learning. *Proceedings of the VLDB Endowment* 16, 7 (2023), 1726–1739. <https://doi.org/10.14778/3587136.3587146>
- [7] Ammar Hayat et al. 2025. A Context-Aware Approach for Enhancing Data Imputation with Pre-trained Language Models. In *Proceedings of the 31st International Conference on Computational Linguistics*.
- [8] X. Huang and S. Hao. 2024. Missing Value Imputation via Pre-trained Language Models with Trainable Prompt and Retrieval Augmentation. *VLDB* (2024).
- [9] Alistair Johnson, Lucas Bulgarelli, Tom Pollard, Brian Gow, Benjamin Moody, Steven Horng, Leo Anthony Celi, and Roger Mark. 2024. MIMIC-IV. *PhysioNet* (Oct. 2024). <https://doi.org/10.13026/kpb9-mt58> Version 3.1.
- [10] Mohamad Kayali et al. 2024. Chorus: Foundation Models for Unified Data Discovery and Exploration. *Proceedings of the VLDB Endowment* 17, 8 (2024), 2104–2114.
- [11] Linkage Library. 2020. Restaurants (Fodors-Zagats), Augmented Version, Fixed Splits. <https://linkagelibrary.icpsr.umich.edu/linkagelibrary/project/127242/view>. Accessed: 2026-05-22.
- [12] Roderick J. A. Little and Donald B. Rubin. 2002. *Statistical Analysis with Missing Data* (2 ed.). Wiley, New York. <https://doi.org/10.1002/9781119013563>
- [13] Mohammad Mahdavi and Ziawasch Abedjan. 2020. Baran: Effective Error Correction via a Unified Context Representation and Transfer Learning. *Proceedings of the VLDB Endowment* 13, 11 (2020), 1948–1961. <https://doi.org/10.14778/3407790.3407801>
- [14] A. Narayan, I. Chami, L. Orr, and C. Ré. 2022. Can Foundation Models Wrangle Your Data? *Proc. VLDB Endow.* 16, 4 (December 2022), 738–746. <https://doi.org/10.14778/3574245.3574258>
- [15] National Library of Medicine. 2026. RxNorm. <https://www.nlm.nih.gov/research/umls/rxnorm/index.html>. Accessed: 2026-05-22.
- [16] Ralph Peeters et al. 2025. Entity Matching using Large Language Models. In *EDBT*.
- [17] R. Peeters and C. Bizer. 2023. Using ChatGPT for entity matching. In *European Conference on Advances in Databases and Information Systems*. Springer Nature Switzerland, Cham, 221–230.
- [18] Ralph Peeters and Christian Bizer. 2025. Entity Matching using Large Language Models. In *Proceedings of the 28th International Conference on Extending Database Technology*.
- [19] Rifaa Sadegh, Ahmed Ahmed Mohameden, Mohamed Lemine Salihi, and Mohamedade Farouk Nanne. 2025. A survey of missing data imputation techniques: statistical methods, machine learning models, and GAN-based approaches. *International Journal of Artificial Intelligence (IJ-AI)* 14, 4 (2025), 2876–2888.
- [20] Yuan Sui, Mengyu Zhou, Mingjie Zhou, Shi Han, and Dongmei Zhang. 2024. Table Meets LLM: Can Large Language Models Understand Structured Table Data? A Benchmark and Empirical Study. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*.
- [21] Olga Troyanskaya, Michael Cantor, Gavin Sherlock, Pat Brown, Trevor Hastie, Robert Tibshirani, David Botstein, and Russ B. Altman. 2001. Missing Value Estimation Methods for DNA Microarrays. *Bioinformatics* 17, 6 (2001), 520–525. <https://doi.org/10.1093/bioinformatics/17.6.520>
- [22] UCI Machine Learning Repository. 1996. Adult Dataset. <https://archive.ics.uci.edu/ml/datasets/adult>. Accessed: 2026-05-22.
- [23] Uppsala Monitoring Centre. 2026. WHODrug Global. <https://who-umc.org/whodrug/whodrug-global/>. Accessed: 2026-05-22.
- [24] U.S. Food and Drug Administration. 2026. Approved Drug Products with Therapeutic Equivalence Evaluations (Orange Book). <https://www.fda.gov/drugs/drug-approvals-and-databases/approved-drug-products-therapeutic-equivalence-evaluations-orange-book>. Accessed: 2026-05-22.
- [25] Haochen Zhang, Yuyang Dong, Chuan Xiao, and Masafumi Oyamada. 2024. Jellyfish: Instruction-Tuning Local Large Language Models for Data Preprocessing. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.

A Entity Matching

For completeness, we include all entity matching experimental figures across datasets and metrics.

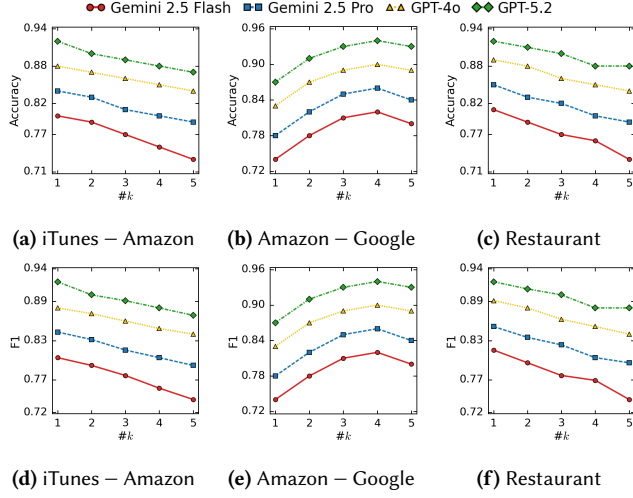


Figure 6: Entity Matching: Vary k (accuracy and F1).

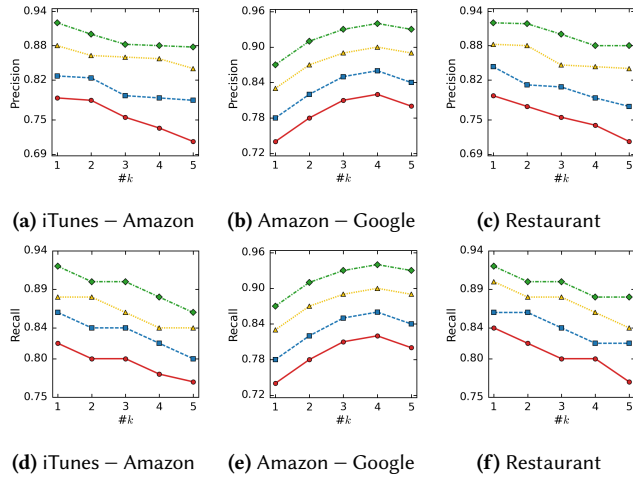


Figure 7: Entity Matching: Vary k (precision and recall).

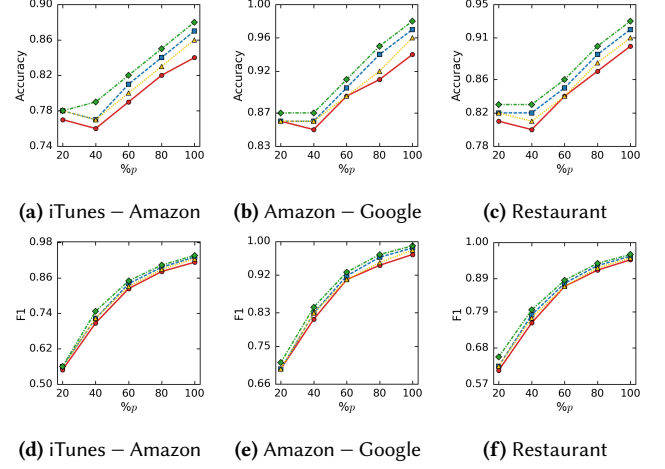


Figure 8: Entity Matching: Vary p (accuracy and F1).

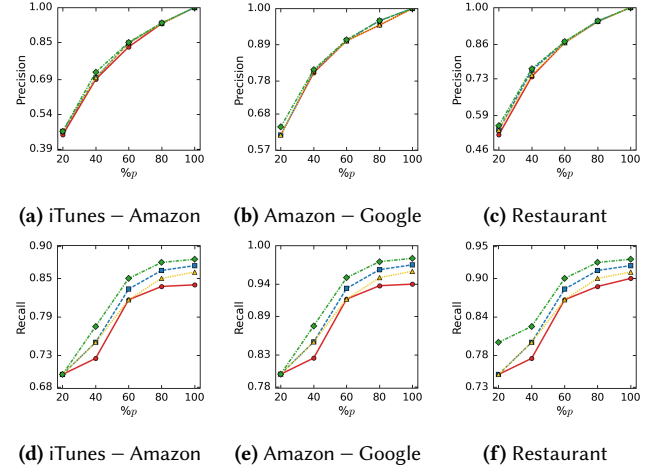


Figure 9: Entity Matching: Vary p (precision and recall).

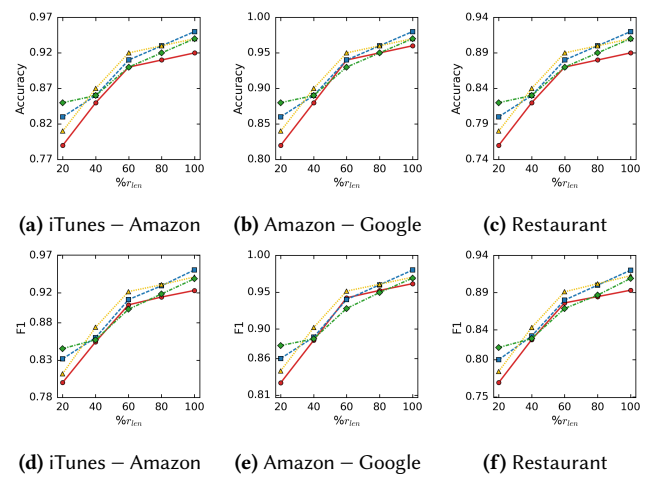
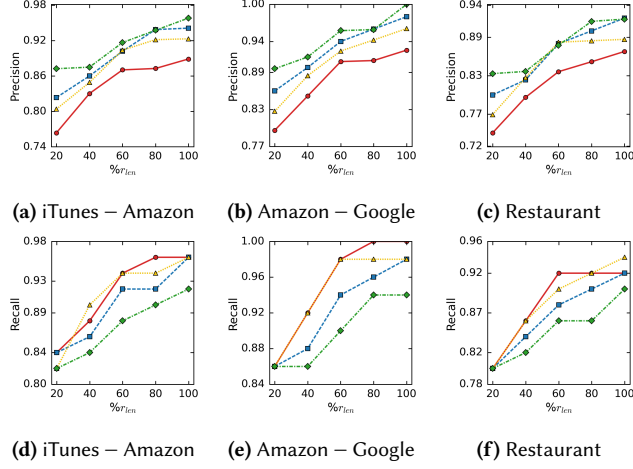
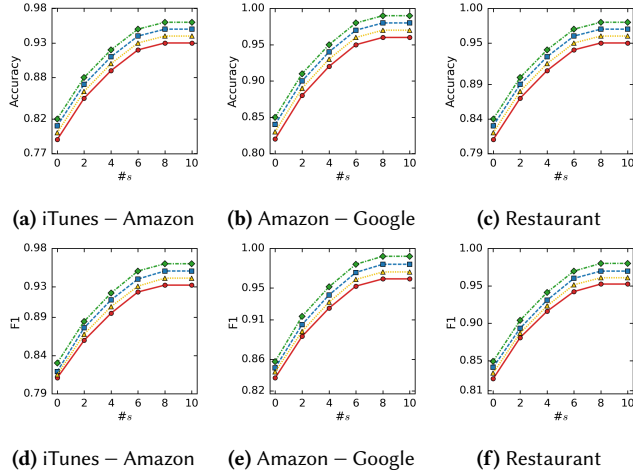
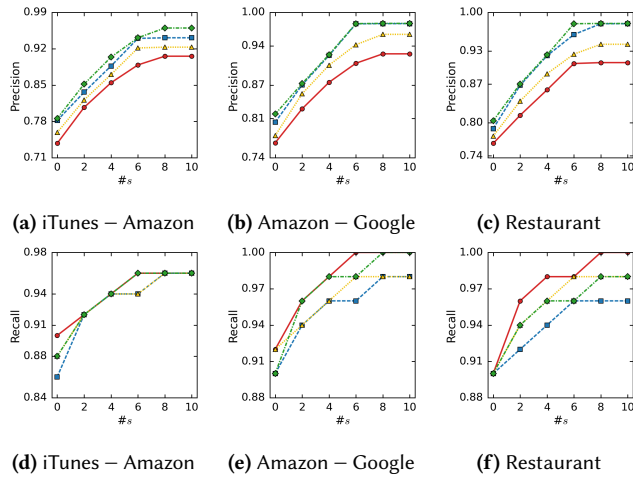
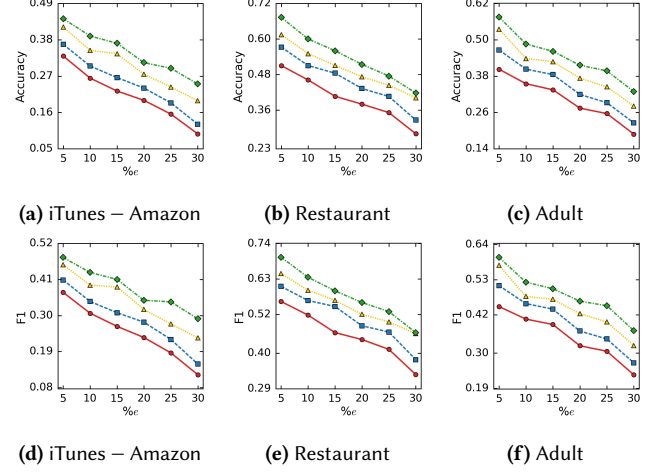
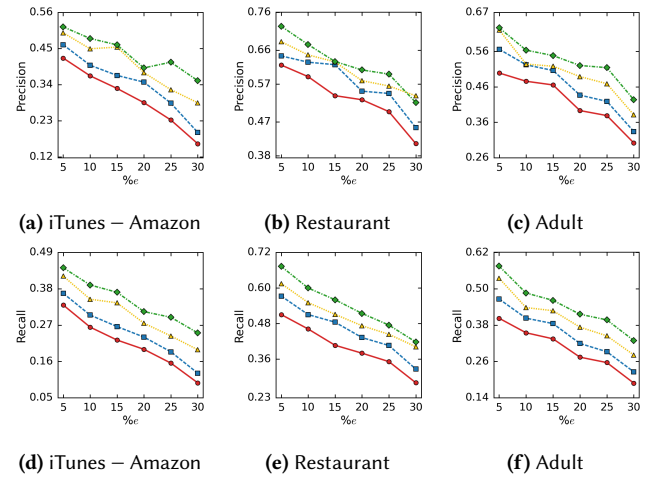


Figure 10: Entity Matching: Vary r_{len} (accuracy and F1).

Figure 11: Entity Matching: Vary r_{len} (precision and recall).Figure 12: Entity Matching: Vary s (accuracy and F1).Figure 13: Entity Matching: Vary s (precision and recall).

B Data Imputation

For completeness, we include all data imputation experimental figures across datasets and metrics.

Figure 14: Data Imputation: Vary e (accuracy and F1).Figure 15: Data Imputation: Vary e (precision and recall).

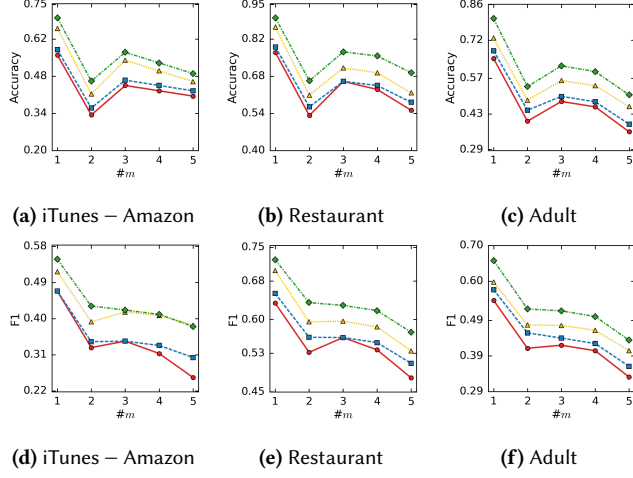


Figure 16: Data Imputation: Vary m (accuracy and F1).

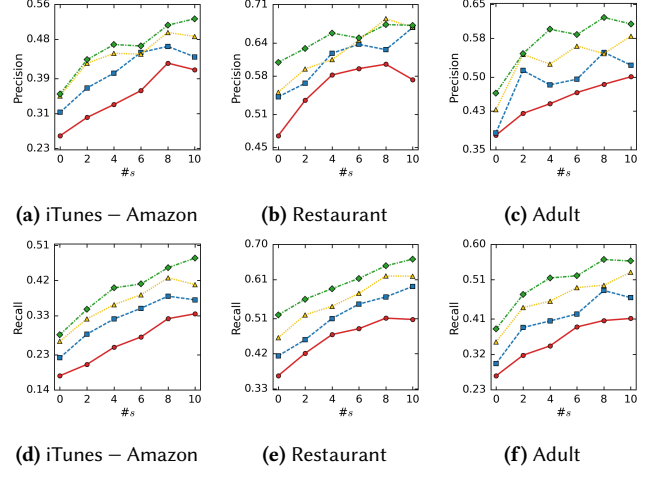


Figure 19: Data Imputation: Vary s (precision and recall).

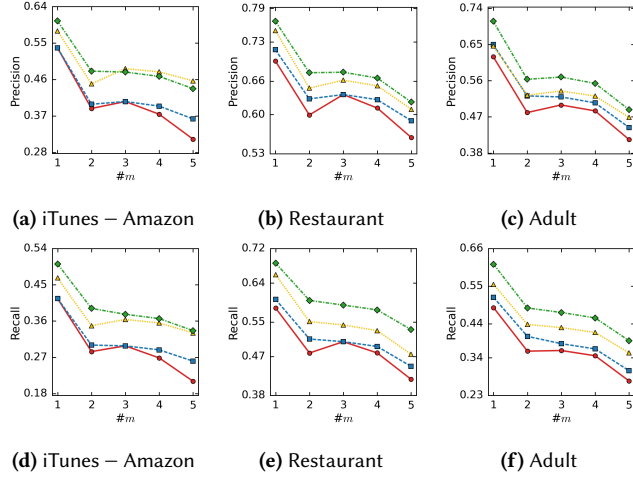


Figure 17: Data Imputation: Vary m (precision and recall).

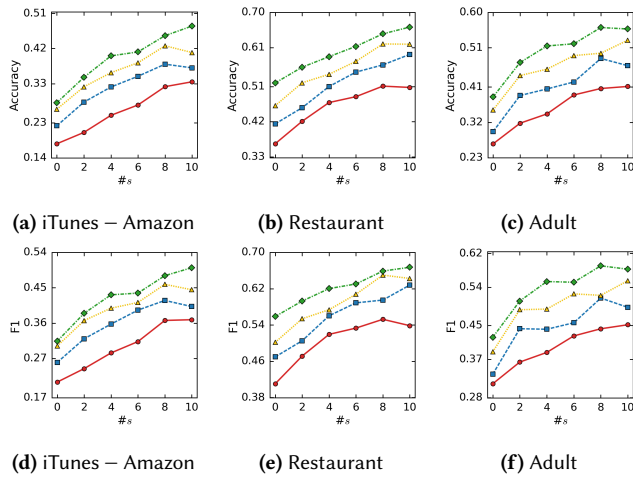


Figure 18: Data Imputation: Vary s (accuracy and F1).

C Semantic Reasoning

For completeness, we include all semantic reasoning experimental figures for the Drug dataset across parameters and metrics.

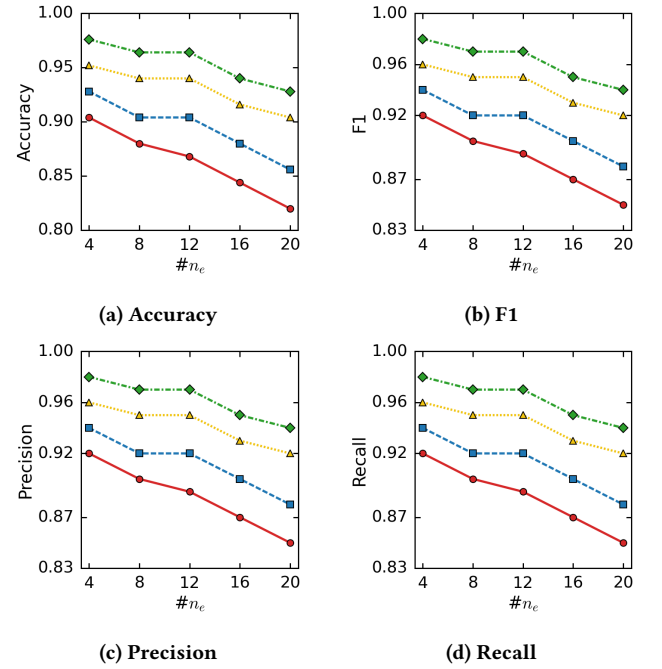
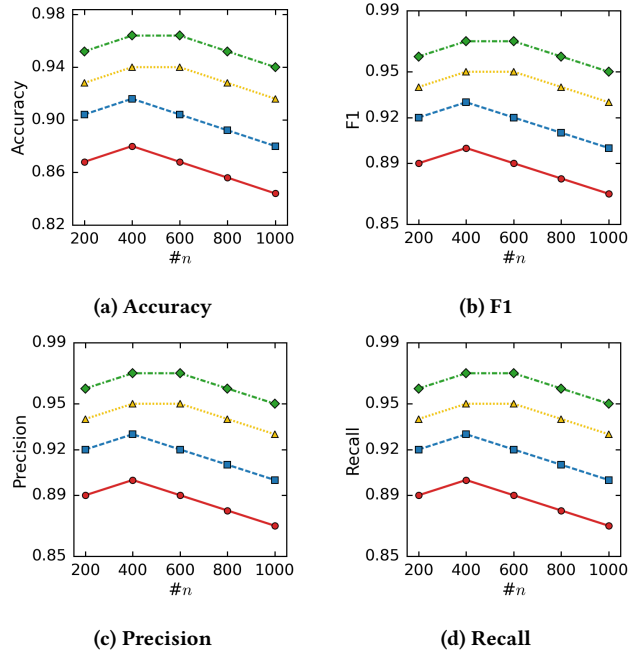
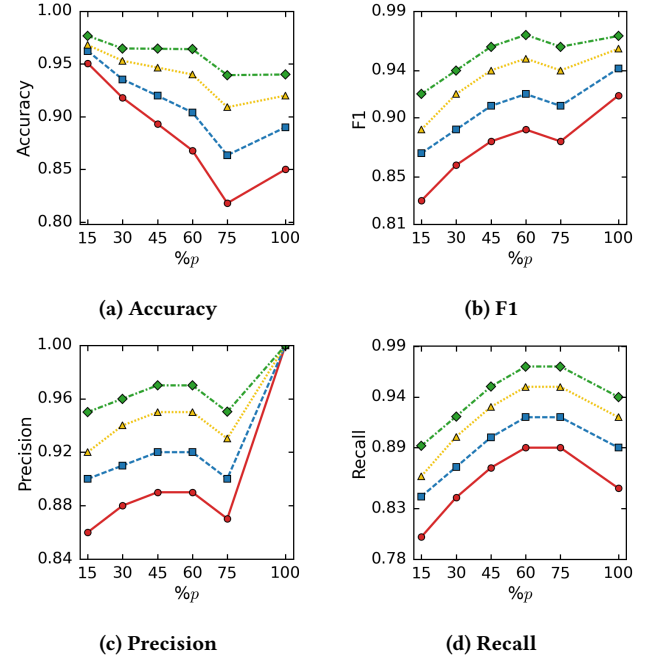


Figure 20: Semantic Reasoning: Vary n_e (Drug).

Figure 21: Semantic Reasoning: Vary n (Drug).Figure 22: Semantic Reasoning: Vary p (Drug).